

SE LAB-1
PES1UG24CS697
Uttam K R

PROBLEM STATEMENT #33

Retail, E-Commerce & Finance

Micro-Lending & Peer Credit Risk Assessor

A peer-to-peer micro-finance platform calculating borrower risk scores using heuristic rules, generating structured repayment schedules, and triggering automated default warnings.

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale (short)	Comments
FR-001	Functional	The system shall generate a structured repayment schedule for loan with specified amounts, due data and balance.	High	Appropriate repayment schedule is generated for a given loan.	The core functional requirement of generating repayment schedules	The repayment schedule should be automatically calculated based on the loan amount, interest rate, and repayment period.

FR-002	Functional	The system shall issue warnings whenever a repayment due is coming up and whenever it is not paid on time as default.	High	Borrower is notified before loan is overdue.	Minimizes payments missed due to borrower forgetting. Core requirement	The system should send reminders before the due date and default warnings when a payment becomes overdue.
FR-003	Functional	The system shall allow the user to view all pending loans and repayment schedules from their profile	Medium	All user loans are shown without error and repayment schedules are listed.	This functionality allows all users to get an overview of their current loans and repayment statuses.	Users should be able to easily track their active loans, upcoming payments, and repayment status from their profile.
FR-004	Functional	The system shall compute borrower credit risk ratings based on debt-to-income ratio, repayment history, and employment stability indicators.	High	Risk tier (Low/Med/High) accurately assigned;	The core peer risk assessment functionality.	The risk rating should be calculated using the borrower's financial and employment information and categorized into Low, Medium, or High risk.
FR-005	Functional	The system shall allow users to submit loan requests with amount, purpose, employment and preferred method of repayment.	High	Loan request is successfully stored in the database.	This is a core functionality as the lenders need to see all the borrowers' requests.	The loan request should validate all required borrower and loan details

						before storing the request for lenders to review.
NFR-001	Nonfunc.	Performance: The risk assessment will be performed and a result will be shown within 3 seconds.	High	All of the risk assessment is performed within a maximum of 3 seconds and is accurate.	Allows fast feedback to the user and increases responsiveness.	The 3-second response requirement should be verified through performance testing under normal and simulated peak usage.
NFR-001	Nonfunc.	Security: The system will secure all borrower and lender details only allowing required people to view them.	High	Unauthorized users cannot access restricted data.	Secures sensitive financial information	Role-based access should ensure borrowers and lenders can access only the information permitted for their role.